GEOMETRY · TRIANGLE RELATIONSHIPS

Triangle Explorer

Explore & Reflect **TEACHER KEY**

Name _____ Date _____ Period _____

Open **Triangle Explorer** on Mr. K's Classroom: Geometry → Triangle Relationships → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

1 Predict before you touch the tool

1. Do you think the angle sum changes when a triangle becomes obtuse?

   No. It is always 180°.

2 Explore use the tool and record what you see

Angles (A(–5, –3), B(5, –3), and C as given)

Drag C to each point. Turn on the exterior angle at C.

C	∠A	∠B	∠C	Sum	Type	Exterior angle at C
(1, 5)	**53.1°**	**63.4°**	**63.4°**	**180°**	**acute**	**116.6° = ∠A + ∠B**
(5, 5)	**38.7°**	**90°**	**51.3°**	**180°**	**right**	**128.7° = ∠A + ∠B**
(–8, –1)	**146.3°**	**8.7°**	**24.9°**	**180°**	**obtuse**	**155.1° = ∠A + ∠B**

Triangle centers (use the obtuse triangle)

Center	Where these lines meet	Inside or outside?
Centroid	**medians**	**inside**
Incenter	**angle bisectors**	**inside**
Circumcenter	**perpendicular bisectors**	**outside**
Orthocenter	**altitudes**	**outside**

Can it be a triangle?

Use the second tab.

Side lengths	Two shortest add to	Triangle?
3, 4, 6	**7 > 6**	**yes**
3, 4, 8	**7 ≤ 8**	**no**
5, 5, 10	**10 ≤ 10**	**no**
4, 6, 9	**10 > 9**	**yes**

3 Write the rule fill in each blank from what you observed

1. The angles of every triangle add up to **180°**.
2. An exterior angle equals the sum of the two **remote** interior angles.
3. The longest side is across from the **largest** angle.

4. The sum of any two sides must be **greater than** the third side.

5. The circumcenter is the same distance from all three **vertices**. The incenter is the same distance from all three **sides**.

4 Check no tool: show your work

1. Two angles are 47° and 68°. Find the third.
   65°
2. The exterior angle at C is 115° and ∠A = 40°. Find ∠B.
   75°
3. Two sides are 5 and 9. What lengths can the third side have?
   Between 4 and 14
4. ∠A = 50°, ∠B = 70°, ∠C = 60°. List the sides from shortest to longest.
   BC, AB, AC

5 Reflect

Why can't 3, 4 and 8 make a triangle?

The 3 and 4 sides together reach only 7, so they can't meet over a side of length 8.